

UNIT: I
ACOUSTICS AND NDT

Pondicherry University 2 marks Question and Answer

1. What do you mean by absorption co-efficient? (Jan 2013, Jan11)

The absorption co-efficient of a material is defined as the ratio of sound energy absorbed by it's surface to that of the total sound energy incident on the surface.

2. Suggest any two ways of reducing the time of reverberation?(Jan 2013)

(i) By providing many windows and ventilators (ii) Using heavy curtains with folds

3. What do you mean by attenuation of ultrasonic Waves? (May 2012, April/May2011)

If ultrasonic waves passed through a medium the measure of intensity per unit time per unit area perpendicular to the direction of propagation decreases with distance from source. This phenomenon is known as attenuation of ultrasonic Waves

4. Define Reverberation time (Jan11 , Jan14, Jan 16, April 10, April / May 2011, 2012, 2015, 2016, 2017)

The persistence of audible sound, even after the source has stopped to emit the sound is called reverberation. The time during which the sound persists in the hall is called as reverberation time.

5. What is Piezo Electric Effect? (Jan 2014, May- 2012, 2014)

When pressure or mechanical force is applied along certain axis (Mechanical axis) with respect to optic axis of the crystal like quartz, tourmaline, rochelle salts etc. than equal and opposite charges are produced along the axis (electrical axis) perpendicular to the optic axis of the crystal. This effect is called Piezoelectric effect.

6. What is meant by Acoustics of Buildings? (Jan10, Jan11, May12)

Acoustics is a branch of Physics that deals with design, generation, detection, and propagation of sound waves in buildings is called acoustics of buildings.

7. What is magnetostriction effect? (May-2013, 2015, 2016, Nov 2017, Sep 2020)

Magnetostriction effect is the principle of producing ultrasonic waves. (ie.) when an alternating magnetic field is applied to a rod of ferromagnetic material such

as nickel, iron, cobalt etc., or alloys of it, then the rod is thrown into longitudinal vibrations, thereby producing ultrasonic waves at resonance.

8. What is meant by Echo? (May2013)

When a reflecting surface is far away from the source (more than 17m) the sound is reflected back as a distinct repetition of direct sound, the reflected sound is called an echo

9. What are transverse ultrasonic waves? (Jan 2010, Jan11, May 2017)

Ultrasonic waves having vibrations perpendicular to the direction of propagation are called as transverse ultrasonic waves.

10. What is Acoustic grating?(Or)How is acoustic grating formed? (Jan 2014, Jan15, May 2018)

When ultrasonic waves are passed through liquid, the density of the liquid varies layer by layer due to the variation in pressure. This periodic variation in density results in the change of refractive index of the liquid and hence the liquid act as a diffraction grating so called acoustic grating. Under this condition when a monochromatic source of light is passed through the acoustical grating, the light gets diffracted.

11. List out any four factors affecting the acoustic quality of a building. (Jan14)

Reverberation Time, Loudness, Focusing, Echoes, Echelon Effect, Resonance, Noises

12. What are ultrasonic waves? (April 2010, May13,Nov2017)

Sound waves having frequencies greater than 20KHz are called transverse ultrasonic waves.

13. What is Echelon effect? (Jan 2015)

If there is a regular repetition of echoes of the original sound to the observer then the effect is called as Echelon effect

14. What is the main difference in the quality of ultrasonic waves produced by piezoelectric and magnetostriction methods? (Jan15)

Using Magneto striction method, frequencies ranging from few hundred Hertz to about 300 KHz can be produced. Using Piezo- Electric effect, Frequencies as high as 500 MHz can be produced.

**15. Mention the name of few universally accepted common methods of NDT.
(Jan-2010, 2015)**

There are two types of methods of NDT. They are,

- i) Pulse echo method (Ultrasonic flaw detector)
- ii) Liquid penetrating method

16. What are X-Cut and Y-Cut crystals. (Jan2013)

X-Cut Crystal: When the crystal is cut perpendicular to the X- axis then it is called X-Cut crystal.

Y-Cut Crystal: When the crystal is cut perpendicular to the Y-axis then it is called Y-Cut Crystal.

17. Mention some the uses of ultrasonic waves. (Jan2014) (Read any 2 points)

- a) They are used in welding and soldering
- b) They are used for cleaning cloths and tiny parts of complicated machines
- c) To detect flaws in materials
- d) For the measurement of the depth of the sea
- e) Ultrasonics are used for detecting tumors and other defects in human body
- f) It is used to find the fetal heart movement and blood circulation
- g) It is used to find the velocity of blood flow
- h) Ultrasonics are used to remove kidney stones and brain tumors without any loss of blood.
- i) Ultrasonic therapy can be used to treat disease like neuralgic and rheumatic pains etc.,

18. Mention any four characteristic features required for the selection of penetrant used in liquid penetrant method.(Jan2012).

- (i) High fluidity
- (ii) Lower viscosity
- (iii) Higher wetability
- (iv) Non- toxic
- (v) Cheap

19. What is standard reverberation time? (or) What is Sabines's law? Jan 2016)

Sabines's law states that the standard reverberation time is the time taken by the sound to fall from one millionth of its original intensity, after the source of sound is stopped

$$I = I_0 / 10^6$$

20. What is Loudness? Give the relation between loudness and intensity of sound.(or) Define loudness and mention the scales of loudness measurement. (May 2017,Sep2019)

Loudness is the degree of sensation produced on the ear. This cannot be measured directly. So that it is measured in terms of intensity. Loudness is proportional to the logarithmic value of intensity

$$L = \log I$$

$L = K \log I$ this is also known as Weber-Fechner's Law.

21. What are sound absorbing materials?(May 2017)

Materials which have the ability to absorb sound waves to provide good quality of sound are called as sound absorbing materials. Examples Carpets, Open window, felt, ordinary chair, glass wooden chairs etc.

**22.What is the basic principle of ultrasonic pulse echo method? (May 2017)
(Or)**

What is Pulse Echo method? (Nov 2016)

Whenever there is a change in medium, the ultrasonic waves will be reflected, is the principle used in ultrasonic flaw detector. Thus, from the intensity of the reflected echoes, the flaws are detected without destroying the material and hence this method is known as a non-destructive testing method.

23. Define Doppler effect.(Jan 2015,May2016,Nov2018)

There is an apparent change in frequency of the sound waves emitted from the source when there is a relative motion between the source and observer. This effect is called Doppler Effect and the shift in frequency is called Doppler shift.

24.What is meant by Non-Destructive testing (NDT)?(May 2018)

Without damaging the component, if it is tested than the testing method is called as non-destructive testing (NDT) method.

25. What are the advantages of a non-destructive testing of products?(May2018)

- They provide the information on the quality of the material or component without damaging.
- Presence of defects can be detected both at the manufacturing stage as well as during service.
- This helps to prevent failures in service and results in not only financial saving but saving human lives also.

26. What is Reverberation? (May2014)

The existence or prolongation or persistence of sound in the hall even after original source is stopped is called **Reverberation (or) lingering** of sound.

27. Write down the condition for resonance in Magnetostriction oscillator? (Jan 2016)

Frequency of the oscillatory circuit = Frequency of the vibrating rod

$$\frac{1}{2\pi\sqrt{L_1C_1}} = \frac{1}{2l}\sqrt{\frac{E}{\rho}}$$

l = Length of the rod

E = Young's modulus of the rod

ρ = Density of the materials of the rod

28 Give the importance of Sabine's law for good auditorium? (May 2014)

Quality of sound is controlled by the reverberation time.

29. What are the advantages and drawback of a Magnetostriction Oscillator? (Nov 16)

ADVANTAGES

- Oscillator construction is very simple and its production cost is low.
- At low frequencies larger power output is possible without the risk of damage to the oscillatory circuit

DISADVANTAGES

- It can produce frequencies up to 3 MHz only
- It is not possible to get a constant single frequency because it depends on the temperature and the degree of magnetization.
- Due to the presence of magnetic field, there will be loss of energy because of hysteresis and eddy current.

30. Explain the properties of ultrasonic waves.(or) What are the properties of ultrasonic waves? (Sep 2019)

- (a) The frequency of an ultrasonic wave is greater than 20,000 Hz.
- (b) It travels longer distance in the medium without any loss.
- (c) When an object is exposed to ultrasonics for a longer time it produces heating effect.
- (d) It has a many modes of vibrations such as longitudinal, transverse and different modes of surface vibrations.
- (e) It undergoes reflection and refraction at the interface due to the change in elastic and physical properties of the medium.

31. What is NDT method and mention any two application of it? (Nov 2018, May 2018)

Important NDT methods are

- a. Liquid penetrant method
- b. Ultrasonic flaw detector
- c. X-ray radiography
- d. Thermography

Application

- 1. The system is used in the aerospace industries for the quality control of production and regular maintenance and safety checks.
- 2. To detect flaws in material
- 3. For the measurement of the depth of the sea.

32. What are the important applications of Radars? (Sep 2020)

- 1. Radar in weather forecasting
- 2. Radar in law enforcement

33. Why ultrasonic frequencies greater than 3MHz cannot be produced by magnetostriction method? (Nov 2018)

- a) It is not possible to get a constant single frequency because it depends on the temperature and the degree of magnetization.
- b) Due to the presence of magnetic field, there will be loss of energy because of hysteresis and eddy current.

PROBLEMS

1. Calculate the reverberation time of a hall of volume 15000m^3 . The average absorption coefficient of absorbing surfaces is 0.3 sabine/m^2 . The total absorbing surface area is 24000 m^2 . (May 2017)

Given:

$$\text{Volume of hall, } V = 15000 \text{ m}^3$$

$$\text{Absorption coefficient, } a = 0.3 \text{ sabine/m}^2$$

$$\text{surface area, } s = 24000 \text{ m}^2$$

$$\text{Reverberation time, } T = \frac{0.167V}{as} = \frac{0.167 \times 15000}{0.3 \times 24000} = 0.3479 \text{ sec}$$

2. A quartz crystal of thickness 0.001m vibrates in its fundamental frequency. Calculate its frequency .Given that $E = 7.9 \times 10^{10} \text{ N/m}^2$ and $\rho = 2650 \text{ Kgm}^{-3}$ for quartz (Jan2011)

Given:

$$\text{Thickness, } t = 1 \times 10^{-3} \text{ m}$$

$$\text{For fundamental mode, } P = 1$$

$$\text{Young's modulus, } E = 7.9 \times 10^{10} \text{ N/m}^2$$

$$\text{Density of the material, } \rho = 2650 \text{ Kgm}^{-3}$$

$$\begin{aligned} \text{The frequency of vibration is given by } f &= \frac{p}{2t} \sqrt{\frac{E}{\rho}} = \frac{1}{2 \times 10^{-3}} \sqrt{\frac{7.9 \times 10^{10}}{2650}} \\ &= 2.7299 \times 10^6 \text{ Hz} \end{aligned}$$

3. Calculate the velocity of ultrasonic waves in a liquid in an acoustic grating using the following data. Wavelength of light used $= 600 \text{ nm}$, frequency of ultrasonic waves $= 100 \text{ MHz}$, Angle of diffraction $= 5^\circ$. (May 18)

Given data:

$$\text{Wavelength of light} = 600 \times 10^{-9} \text{ m}$$

$$\text{Frequency of ultrasonic waves} = 100 \text{ MHz} = 100 \times 10^6 \text{ Hz}$$

$$\text{Velocity of ultrasonic waves in a liquid } V = v_u n \lambda_w \sin \theta$$

$$= (100 \times 10^6 \times 600 \times 10^{-9}) / \sin 5^\circ \text{ (for first order } n = 1)$$

$$\text{Velocity of ultrasonic waves in a liquid } V = 688.0 \text{ m/s}$$

Pondicherry University (Part – B 11 marks Questions & Answer)

1. Give a detailed account on the basic principle, the construction and production of ultrasonic waves using Magnetostriction oscillator.

(Jan 10, Jan 11, April 11, Jan 13, Jan 14, Jan15, Nov16, Jan 16, May 2012,2014,2017,Nov/Dec 2018)

PRINCIPLE :

MAGNETOSTRICTION EFFECT :

The magnetostriction effect is the principle of producing ultrasonic waves.(i.e.)When an alternating magnetic field is applied to a rod of ferromagnetic material such as nickel, iron, cobalt etc., alloys of it, then the rod is thrown into longitudinal vibrations thereby producing ultrasonic waves at resonance. This is known as *magnetostriction effect*.

CONSTRUCTION:

AB is a ferromagnetic (nickel) rod clamped at the middle. L_1 and L_2 are two coils wound on the ends of the rod to set up magnetic field. The inductor coil L_1 and the variable condenser C_1 form the tank circuit.

This tank circuit is connected in between the collector and the emitter of an n-p-n transistor as shown in Fig. 1.2. Coil L_2 is connected to the base to feed back supply. The ammeter connected to see the current flow in the circuit. The input to the transistor is taken from the power supply.

WORKING:

The battery is switched ON and hence current is produced by the transistor. The current is passed through the coil L_1 , which causes a corresponding change in the magnetization of the rod. Now the rod starts vibrating due to magnetostriction effect. We know that when a coil is wound over a vibrating nickel rod, an emf is induced in the coil called **inverse magnetostriction effect**.

The induced e.m.f. is fed to the base of the transistor which acts as a feedback continuously. In this way, the current in the transistor, is built up and the vibrations of the rod is maintained. The frequency of the oscillatory circuit is adjusted by the capacitor C_1 . When this frequency is equal to the natural frequency of the vibrating rod, resonance occurs.

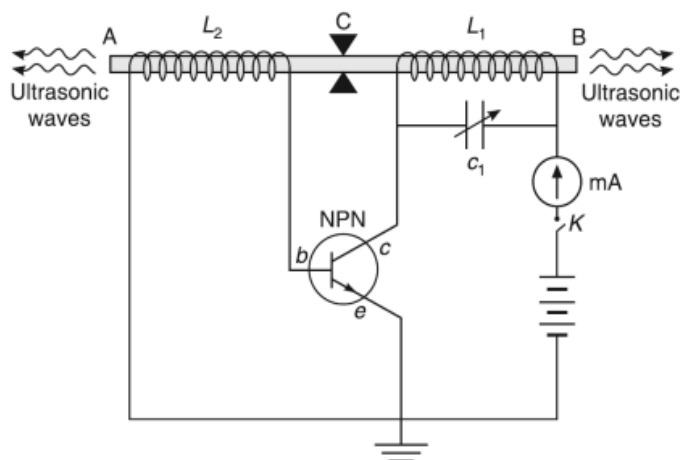


Fig.1.2. Magnetostriction generator

At resonance the rod vibrates longitudinally with larger amplitude, thereby producing ultrasonic waves of high frequency along both the ends of the rod. By adjusting the length of the rod and the capacity of the condenser for resonance, high frequencies (i.e.) oscillations of different frequency can be obtained.

Condition for resonance:

Frequency of the oscillatory circuit = Frequency of the vibrating rod

$$\frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{1}{2l} \sqrt{\frac{E}{\rho}}$$

l = Length of the rod

E = Young's modulus of the rod

ρ = Density of the materials of the rod

2. Write an essay about the acoustics of buildings and the factors affecting it.

(Jan 13, Jan 14, April 10, May 11,12,14,15, Jan 16, Nov17)

The factors affecting the acoustics (sound) of buildings are as follows :

- (i) Unoptimised reverberation time.
- (ii) Very low (or) very high loudness.
- (iii) Improper focussing of sound to a particular area, which may cause interference.
- (iv) Echoes (or) Echelon effects produced inside the buildings.
- (v) Resonance caused due to matching of sound waves.

- (vi) Unwanted sound from outside (or) inside the building, so called noise may also affect the acoustical property of buildings.

(i) Unoptimised Reverberation time and its remedy :

We know reverberation time is the time taken for the sound to fall one millionth of its original sound intensity, when source is switched off. This reverberation time plays a vital role in the auditorium, for clear audibility of sound.

If the reverberation time is very high then it produces, echoes in the hall and if the reverberation time is very low, the sound will not be clearly heard by the audience. Therefore for clear audibility, we should maintain optimum reverberation time. The optimum reverberation time can be achieved by the following steps.

1. By having the full capacity of audience in the auditorium.
2. By choosing absorbents like felt, fibre, board, glass etc. inside the auditorium and even at the back of chairs.
3. Reverberation time can be optimised by providing windows and ventilators at the places wherever necessary and using curtains with folds for the window.
4. The reverberation time can also be optimised by decorating the walls with beautiful pictures.

(ii) Loudness

We know loudness is the degree of sensation produced on the ear, it varies from observer to observer. But, it is found that for a single observer the loudness varies from one place to another in the same auditorium. This defect is caused due to the bad acoustical construction of buildings. The loudness will be very low in some area and will be very high in some areas. It can be optimized by the following remedies.

Remedies :

- a. Loudspeakers should be placed at the places where we have low loudness.
- b. The loudness can also be increased by making reflecting surfaces, wherever necessary.
- c. Loudness can be increased by constructing low ceilings.
- d. Absorbents are placed at the places where we have high loudness.

Thus, the loudness should be made even, all over the auditorium, so that the observer can hear the sound at a constant loudness at all the places.

(iii) Focussing and Interference Effects :

In some places of a hall, the sound will not be heard properly and that place is said to be a dead space, which is due to the presence of convex or concave surfaces in the hall as shown in the below figure 1.7.

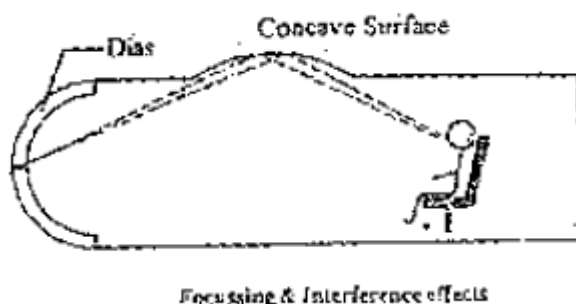


Fig.1.7.

Sometimes the sound waves will have interference pattern because of the ceiling surfaces which will create maximum intensity of sound (due to constructive interference) in some places and minimum intensity of sound (due to destructive interference) at some other places and hence causing uneven distribution of sound intensity in the hall. These problems can be rectified by the following ways.

Remedies :

1. By avoiding curved surface (or) covering the curved surfaces by suitable absorbents, the focusing can be avoided.
2. By evenly polishing and decorating with absorbents the interference effect can be avoided.

(iv) Echoes and Echelon effect :

In some halls, the walls of the halls will scatter the sound waves rather than reflecting it, this may create nuisance effect due to echoes. The echoes are formed when the time interval between the direct and reflected sound waves are about $\frac{1}{15}$ th of a second. This effect occurs due to the reason that the reflected sound waves reaches the observer later than the direct sound. If there is a regular repetition of echoes of the original sound to the observer then the effect is called as **Echelon effect** (eg.) sound produced from equally spaced steps in a stair case at regular intervals produces echelon effect.

Remedy:

The echo can be avoided by lining the surfaces with suitable sound absorbing materials and by providing enough number of doors and windows.

(v) Resonance:

Resonance occurs when a new sound note of frequency matches with standard audio frequency. Sometimes the window - panel, sections of the wooden portion are thrown into vibrations to produce new sounds, which results in interference between original sounds and create sound. This will create disturbance to the audience.

Remedies:

1. The resonance effect can be avoided by providing proper ventilation and by adjusting the reverberation time to the optimum level.
2. Nowadays the resonance is completely eliminated by air conditioning the halls.

(vi) Noise :

Noise is an unwanted sound produce due to heavy traffic outside the hall which leads to displeasing effect on the ear. There are three types of noises.

1. Air borne noise.
2. Structure borne noise.
3. Inside noise

All these three noises pollute the area at which it has been produced and create harmful effects to the human beings. Fortunately, human beings have the capability to reject the sound within certain limits with conscious efforts and to carry on his normal work. But sometimes the noises are strong which results in the following effects.

Effects produced due to noise pollution:

- i. It produces mental fatigue and irritation.
- ii. It diverts the concentration on work and hence reduces the efficiency of the work.
- iii. Some strong noises lead to damage the eardrum and make the worker hearing impaired.
- iv. The noises which are produced regularly will even retard the normal growth of infants and young children.

(i). Air borne noise :

The noise which reaches the hall through open window, doors and ventilations are called as **air borne noise**. This type of noise is produced both in rural areas [natural sound of wind and animals] and in urban areas [noise that arises from factories, aircrafts, automobiles, trains, flights etc.]

Remedies:

- i. By making the hall air conditioned, this noise may be eliminated.
- ii. By allotting proper places for doors and windows, this noise can be reduced.
- iii. It can be further reduced by using double doors and windows with separate frames and by placing the absorbents in between them.

(ii). Structure borne noise:

The noise that reaches the hall through the structure of the building is termed as structure borne noise. These types of noises are produced inside the building. Which may be due to the machinery operation, movement of furnitures, foot steps etc., and these sound will produce structural vibration giving rise to structure borne noise.

Remedies:

- i. By properly breaking the continuity of the interposing layers by some acoustical insulators this type of noise can be avoided.
- ii. By providing carpets, resilient, antivibration mounts etc., this type of noise can be reduced.

(iii) Inside noise:

The noises that are produced inside the halls is known as **inside noise**. For example in some offices the sound produced by machinery, type writers etc., and produces this type of noise.

Remedies :

- i. By placing the machineries and type writers over the absorbing materials.
- ii. It can be reduced by covering the floors with the carpet.
- iii. By fitting the engine on the floor with a layer of wood (or) felt between them this type of noise can be avoided.

3. Explain with neat sketch the construction and production of ultrasonic waves using piezoelectric oscillator. (Jan 14, 16 May 11, 13, 15, 16, 17, 18, Nov17,Sep2020, April2018, Nov2018)

CONSTRUCTION:

A thin slice of quartz crystal Q is placed in between two metal plates A and B. The plates A and B are connected to the coil L₃. The coils L₁, L₂ and L₃ are inductively coupled as shown in Fig.1.4.

The coil L₁ is connected to base of the transistor circuit and coil L₂ is connected to the collector of the transistor. The capacitor C₁ is used to vary the frequency of the oscillatory circuit (L₁C₁).

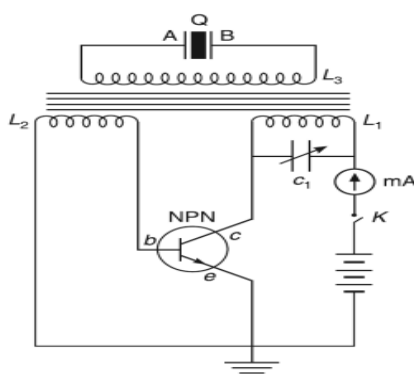


Fig.1.4. Piezoelectric generator

WORKING :

When the battery is switched ON, the current is passed through the coils L₁ and L₂ of the primary circuit. The base circuit produces alternating voltage with

$$\text{frequency } f = \frac{1}{2\pi\sqrt{L_1 C_1}}$$

Due to transformer action, the current is transferred to the secondary circuit and fed to the electrodes A and B. Now the crystal is under high frequency alternating voltage. Due to inverse piezo – electric effect, the crystal starts vibrating along the mechanical axis of the crystal. The frequency of the base circuit is adjusted by the variable capacitor C₁. **When this frequency is equal to the natural frequency of the vibrating crystal, resonance occurs.** At resonance the crystal vibrates longitudinally with larger amplitude, thereby producing ultrasonic waves of high frequency along both the ends of the crystal.

Condition for resonance

Frequency of the oscillatory circuit = Frequency of the vibrating crystal

$$\frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{P}{2l} \sqrt{\frac{E}{\rho}} \quad P = 1, 2, 3 (\text{Modes of vibration})$$

l - length of the crystal

E - Young's modulus of the crystal

ρ - density of the crystal

$P = 1, 2, 3$ (Modes of vibration)

4. Derive Sabine's mathematical relation for reverberation time

(Jan 10, 11, 14, 15, May 13, 16, 17, 18, Nov 16, 18, Sep 2020, 2019)

The relation connecting the reverberation time with the volume of the hall, the area and absorption co-efficient is known as **Sabine's formula**.

Sabine's developed the formula to express the rise and fall of sound intensity by the following assumptions.

- (i) Distribution of sound energy is uniform throughout the hall.
- (ii) There is Interference between sound waves.
- (iii) The absorption co-efficient is independent of sound intensity.
- (iv) The Rate of emission of sound energy from the source is constant.

Let us consider a small element ' ds ' on a plane wall AB . Assume that the element ' ds ' receives the sound energy ' E '. Let us draw two concentric circles of radii ' r ' and ' $r + dr$ ' from the centre point ' O ' of ds .

Consider a small shaded portion lying between the two semi circles drawn at an angle ' θ ' and ' $\theta + d\theta$ ', with the normal to ' ds ' as shown in the Figure 1.9. Let ' dr ' be the radial length and ' $rd\theta$ ' be the arc length. Then,

$$\text{Area of the shaded portion} = rd\theta dr \text{ ----- (1)}$$

If the whole figure is rotated about the normal through an angle $d\phi$ (Fig 1.10).

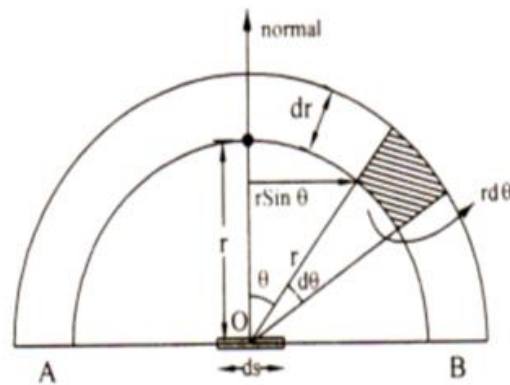


Fig 1.9

Then it is evident the area of the shaded portion travels through a small distance say 'dx'

$$dx = r \sin \theta \cdot d\phi \text{-----(2)}$$

Volume traced by the shaded portion is

$$dV = \text{Area} \times \text{distance}$$

Substituting from equation (1) and (2) in the above equation we have

$$dV = rd\theta dr \cdot r \sin \theta \cdot d\phi$$

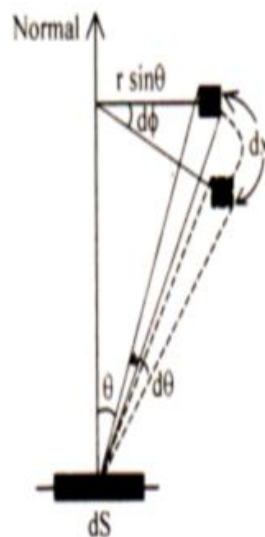


Fig.1.10.

$$dv = r^2 \sin \theta \cdot dr \cdot d\theta \cdot d\phi$$

The sound energy present in this volume dv is = $E \times$ volume of the shaded portion.

$$= E r^2 \sin \theta \cdot dr \cdot d\theta \cdot d\phi$$

This sound energy will travel through the element in all the directions.

The sound energy present in this volume 'dv' per unit solid angle is

$$= \frac{E r^2 \sin\theta .dr.d\theta.d\phi}{4\pi}$$

Therefore in this case the solid angle subtended by the area 'ds' in the figure 1.11. at this element of volume 'dv' is $d\omega = ds'/r^2$

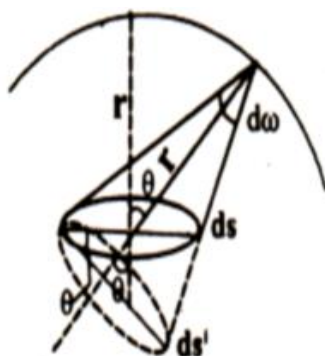


Fig1.11

$$\cos\theta = \frac{ds'}{ds} \text{ (or) } ds' = ds \cos\theta$$

Therefore we can write solid angle subtended by the area 'ds' as

$$d\omega = \frac{ds.\cos\theta}{r^2}$$

Hence, the sound energy travelling from the element (i.e from $d\omega$) towards 'ds'

$$= \frac{E r^2 \sin\theta .dr.d\theta.d\phi}{4\pi} \frac{ds.\cos\theta}{r^2} \dots (3)$$

To find the total energy received by the element 'ds' per second , we have to integrate the eq (3) for the whole volume lying within a distance 'v' of 'ds' .where v is velocity of sound.

It is obvious from the geometry of the figure that,

ϕ changes from 0 to 2π

θ changes from 0 to $\frac{\pi}{2}$

r changes from 0 to v

Integrating equation (3) with respect to these limits we can write energy received per second by ds is

$$ds = \frac{E ds}{4\pi} \int_0^v dr \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta \int_0^{2\pi} d\phi$$

$$\text{since } \int_0^{2\pi} d\phi = 2\pi$$

$$ds = \frac{E ds}{4\pi} \int_0^v dr \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta$$

$$ds = \frac{E ds}{4} \int_0^v dr \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta$$

$$ds = \frac{E ds}{4} \int_0^v dr \int_0^{\frac{\pi}{2}} 2\sin\theta \cos\theta d\theta$$

$$\text{since } 2\sin\theta \cos\theta d\theta = \sin 2\theta$$

$$ds = \frac{E ds}{4} \int_0^v dr \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta \quad (\text{since } \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta = 1)$$

$$= \frac{E \cdot ds \cdot v}{4} \quad \dots(4)$$

Let 'a' be the absorption coefficient of the area ds.

17

$$\therefore \text{Energy absorbed by ds in unit time} = \frac{1}{4} E \cdot v \cdot a \cdot ds$$

$$\therefore \text{Total absorption at all the surfaces of the wall} = \frac{1}{4} E \cdot v \cdot \sum a \cdot ds$$

$$\text{Total rate of energy absorption} = \frac{1}{4} E \cdot v \cdot A \quad \dots(5)$$

Where 'E' is the energy from sources and 'A' is the total absorption on all the surfaces on which the sound falls

$$(\text{ie.}) A = \sum a \cdot ds$$

1.7.3. Growth and Decay of the sound energy

If 'P' is the power output (ie. The rate of emission of sound energy from the source) then we can write

$$(\text{ie}) \text{ Power output } P = \frac{1}{4} E_m \cdot v \cdot A$$

Here E_m is maximum energy from the source (which has been emitted) that is nothing but maximum energy which incidents on the wall

$$\therefore E_m = \frac{4P}{vA} \quad \dots(6)$$

If 'V' is the volume of the hall then we can write the total energy at any instant

$$'t' = EV$$

$$\therefore \text{Rate of growth (or) increase in energy} = \frac{d}{dt}(EV) = V \cdot \frac{dE}{dt} \dots\dots(7)$$

At any instant

Rate of growth of Energy = Rate of supply of energy from the source -

Rate of absorption of energy by walls

$\therefore$ From eq (5) and eq (7) we can write

$$V \cdot \frac{dE}{dt} = P - \frac{1}{4}EvA$$

$$\frac{dE}{dt} = \frac{P}{V} - \frac{EvA}{4V}$$

$$\text{Let } \alpha = \frac{vA}{4V}$$

$$\therefore \frac{dE}{dt} = \frac{4P}{vA} \alpha - E\alpha$$

$$(\text{or}) \frac{dE}{dt} + E\alpha = \frac{4P}{vA} \alpha$$

Multiplying by $e^{\alpha t}$ on both sides , we have

$$\left(\frac{dE}{dt} + E\alpha \right) e^{\alpha t} = \left(\frac{4P}{vA} \right) e^{\alpha t} \cdot \alpha$$

$$(\text{or}) \frac{d}{dt}(Ee^{\alpha t}) = \left(\frac{4P}{vA} \right) \cdot \alpha \cdot e^{\alpha t}$$

Integrating and solving we get

$$Ee^{\alpha t} = \frac{4P}{vA} \cdot e^{\alpha t} + K \dots\dots(8)$$

Where K is a constant of integration

(i) Growth of sound energy:

Let us first evaluate K for growth the boundary conditions are at $t=0$, $E=0$,

$\therefore$ eq. (8) becomes

$$0 = \frac{4P}{vA} \cdot 1 + K (\text{or}) - \frac{4P}{vA} = K$$

Therefore substituting the value of K in eq (8) we can write

$$Ee^{\alpha t} = \frac{4P}{vA} \cdot e^{\alpha t} - \frac{4P}{vA}$$

$$Ee^{\alpha t} = \frac{4P}{vA} (e^{\alpha t} - 1)$$

Since from eq. (6) $E_m = \frac{4P}{vA}$, we can write

$$Ee^{\alpha t} = E_m(e^{\alpha t} - 1)$$

Divide by $e^{\alpha t}$ throughout, we get

$$E = E_m \left(1 - \frac{1}{e^{\alpha t}}\right)$$

$$E = E_m (1 - e^{-\alpha t}) \quad \dots(9)$$

Where E_m is the maximum sound energy. This expression gives the growth of sound energy density 'E' with time 't'. The growth is along an exponential curve as shown in the figure 1.12.

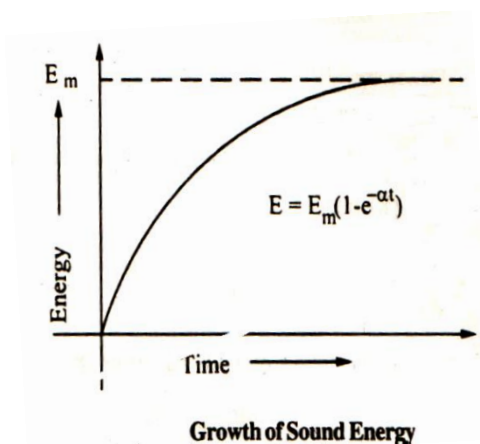


Fig.1.12.

This indicates that E increases until $t = \infty$ at $t = \infty$ $E = E_{max}$

(ii) Decay of sound energy

Let us first evaluate K for decay. Here boundary condition are at $t=0$; $E=E_m$. Initially the sound increases from E to E_m and now it is going to decay from E_m . Therefore time is considered as zero for $E=E_m$ (ie.). At $E = E_m$ the sound energy from the source is cut off. Therefore rate of emission of sound energy from the source = 0 (ie.) $P=0$.

Therefore from eq (8) we can write

$$E_m e^0 = 0 + K \text{ (or) } K = E_m$$

Therefore substituting the value of K for decay in eq (8).we get

$$Ee^{\alpha t} = \frac{4P}{vA} \cdot e^{\alpha t} + E_m$$

Since $P=0$ (ie.)Energy from source is cut off for decay of sound

We can write $Ee^{\alpha t} = E_m$

$$E = E_m e^{-\alpha t} \dots\dots\dots (10)$$

Eq (10) gives the decay of sound energy density with time even after the source is cut off. It is an exponentially decreasing function from maximum energy (E_m) as shown in the below figure 1.13(a) the growth and decay sound energy is also represented in the figure 1.13(b)

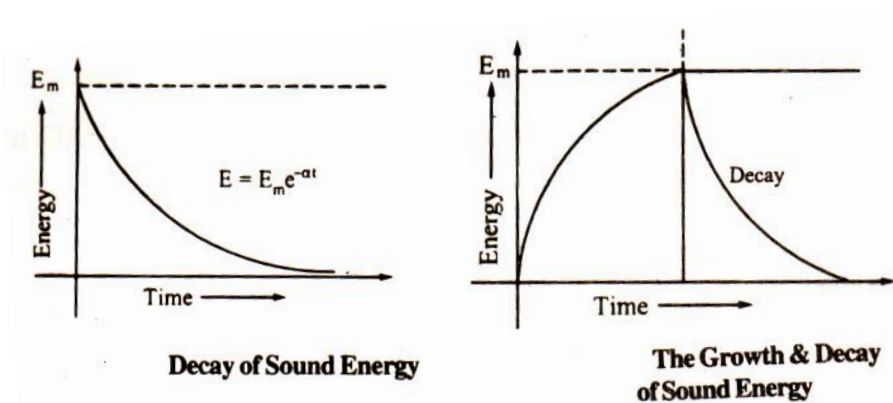


Fig.1.13 (a)

Fig1.13(b)

1.7.4. Proof for Reverberation time (T)

According to Sabine the reverberation time is defined as the time taken by a sound to fall one millionth of its initial value when the source of sound is cut off.

The condition is , at $t=T$; $E = \frac{E_m}{10^6}$

Eq (10) becomes $\frac{E_m}{10^6} = E_m e^{-\alpha t}$ (or) $10^{-6} = e^{-\alpha t}$

Take log on both sides

$$\alpha T = \log e^{10^6}$$

$$\alpha T = 6 \log e^{10}$$

$$\alpha T = 6 \times 2.3026 \log 10^{10}$$

$$\alpha T = 6 \times 2.3026 \times 1 \dots\dots(11)$$

We know that

$$\alpha = \frac{vA}{4V} \dots\dots(12)$$

Substituting eq (12) in eq (11). We get

$$\frac{vA}{4V} T = 6 \times 2.3026$$

$$T = \frac{4V \times 6 \times 2.3026}{vA}$$

We know that velocity of sound in air at room temperature $v=330\text{m/s}$

$$T = \frac{4V \times 6 \times 2.3026}{330 \times A}$$

$$T = \frac{V \times 0.167}{A}$$

$$T = \frac{V \times 0.167}{\sum as} \quad \dots(13)$$

Where $\sum as = a_1s_1 + a_2s_2 + \dots$

Eq (13) represents the reverberation time, which depends on three factors viz.,

- (i) Volume of the hall (v)
- (ii) Surface area (s)
- (iii) Absorption co-efficient (a) of the materials kept inside the hall.

Among these three factors. Volume is fixed. Therefore, the reverberation time can be optimized by either varying the surface area of the reflecting surfaces or the absorption coefficient of the materials used inside the hall.

5. Calculate the frequency of the fundamental note and the first overtone emitted by a piezoelectric crystal, using the following data. Vibrating length = 5mm; young's modulus = $7.9 \times 10^{10} \text{ Nm}^{-2}$ and density of crystal = 2650 kgm^{-3} ? (Jan 11)

Given:

Thickness, t = $1 \times 10^{-3} \text{ m}$

For fundamental mode, P = 1

Young's modulus, E = $7.9 \times 10^{10} \text{ N/m}^2$

Density of the material, ρ = 2650 Kg m^{-3}

$$\begin{aligned} \text{The frequency of vibration is given by } f &= \frac{p}{2t} \sqrt{\frac{E}{\rho}} = \frac{1}{2 \times 10^{-3}} \sqrt{\frac{7.9 \times 10^{10}}{2650}} \\ &= 2.7299 \times 10^6 \text{ Hz} \end{aligned}$$

6. Give the liquid penetrant method of surface analysis (Jan 14, 15, 16, May14, 16 Sep2020)

Liquid penetrant testing is the simplest non-destructive method of finding discontinuities, openings, cracks etc., in the surface of a nonporous solid material.

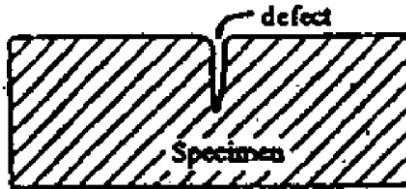


Fig1.16

Principle:

A liquid (penetrant) is applied to a cleaned surface of the specimen and sufficient time is allowed for penetration of the liquid into the defect by capillary action as shown in Fig 1.16. After sufficient time has passed for the penetrant to enter discontinuity, the surface of the part is cleaned. The capillary action is again employed to act as a blotter to draw penetrant from the discontinuity.

Different steps involved in testing

There are five essential steps in the liquid penetrant inspection method. They are

- a) Cleaning
- b) Application of penetrant
- c) Removal of excess penetrant
- d) Application of developer
- e) Inspection and evaluation

a) Cleaning

All surfaces of a specimen must be thoroughly cleaned and completely dried before it is subjected to inspection. To clean the surface, detergent and solvents like petrol or benzene etc., can be used.

b) Application of penetrant

After surface preparation, liquid penetrant is applied in a suitable manner, so as to form a film of penetrant over the component surface as shown in Fig 1.17. The

liquid film should remain on the surface for a sufficient period for full penetration of the penetrant into surface defects.

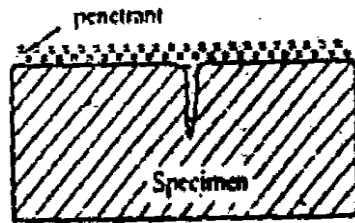


Fig.1.17

c) Removal of excess penetrant



Fig.1.18

It is now necessary to remove excess penetrant from the surface of the component. Uniform removal of excess penetrant is necessary for effective inspection but over cleaning must be avoided (Fig1.18).

d) Application of developer

The development stage is necessary to reveal clearly the presence of defects. A developer usually a very fine chalk powder is applied on the surface under inspection and it forms uniform layer over the surface of the component. The developer acts as a blotter. Penetrant liquid present within defects will be slowly drawn by capillary action into the pores of the chalk. There will be some spread of penetrant within the developer and this will magnify the apparent width of a defect. The developer also provides a uniform background to assist visual inspection (Fig.1.19)

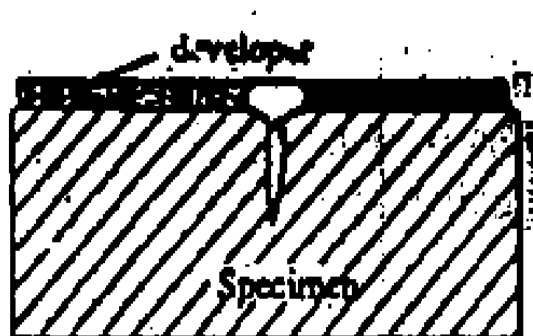


Fig. 1.19

e) Inspection and evaluation

After an optimum developing time has been allowed, the component surface is inspected for indications of penetrant 'bleedback' into the developer. Dye penetrant inspection is carried out in strong lighting conditions, while fluorescent penetrant inspection is performed in a suitable screened area using ultra-violet light. The latter technique causes the penetrant to emit visible light and defects are brilliantly outlined.

Advantages

- a) The liquid penetrant process is simple
- b) The equipment necessary is cheaper
- c) The establishment of procedures and inspection standards for specific product parts is usually less difficult than more sophisticated methods
- d) This technique can be employed for any material except porous materials
- e) Penetrant inspection is suitable for components of virtually any size or shape

Limitations

- a) It can detect surface-breaking defects only
- b) This method is not suitable for rough surfaces and porous materials

7. Give the pulse echo method of ultrasonic waves (Jan13, 14, 16, May 17) (OR)

How are ultrasonic waves used to measure the depth of sea? Explain (Nov 17, Sep2020, Sep2019)

PRINCIPLE :

Whenever there is a change in medium, the ultrasonic waves will be reflected, is the principle used in ultrasonic flaw detector. Thus, from the intensity of the reflected echoes, **the flaws are detected without destroying the material and hence this method is known as a non-destructive testing method.**

DESCRIPTION :

It consists of a Piezo electric transducer coupled to the upper surface of the specimen (metal) without any air gap between the specimen and the transducer. A frequency generator is connected to the transducer to generate high frequency pulses. The total set up is connected to the amplifier and to a cathode ray oscilloscope as shown in the figure 1.14.

- i. The pulse generator generates a high potential difference and is applied to the piezo-electric transducers.
- ii. The piezo-electric crystals are resonated to produce ultrasonic waves.
- iii. These ultrasonic waves (pulse A) are recorded in CRO and is transmitted through the specimen.

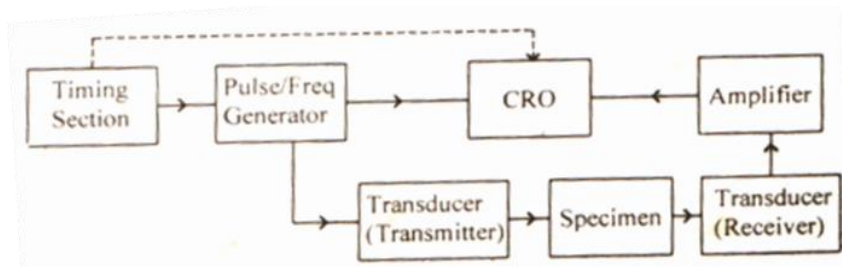


Fig.1.14

- iv. These waves travel through the specimen (metal) and is reflected back by the other end.
- v. The reflected ultrasonics (pulse B) is received by the transducer.
- vi. These reflected signals are amplified and is found to be almost the same as that of the transmitted signals as shown in the figure 1.15 which shows that there is no defect in the specimen.

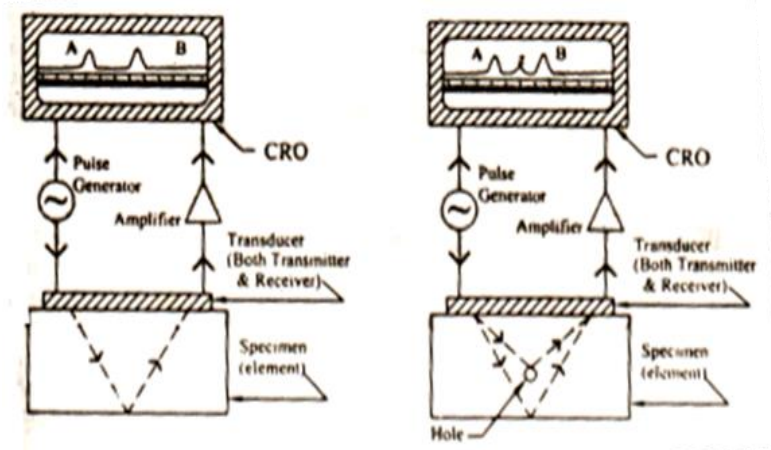


Fig. 1.15

- vii. If there is any defect on the specimen like a small hole (or) pores, then the ultrasonic will be reflected by the holes (ie.) defects due to the change in medium.

- viii. These defects give rise to another signal (pulse Z) in between pulses 'A' and 'B'. Similarly if we have many such holes, many Z - pulses will be seen over the screen of CRO as shown in the figure 1.15.
- ix. From the time delay between the transmitted and received pulses the position of the hole can be found.
- x. From the height of the pulses received the depth of the hole can also be determined.

Advantages:

- 1. This method is highly sensitive to most of the cracks and flaws.
- 2. It gives immediate results at very low cost at a very high speed.
- 3. It indicates the size and location of the flaws exactly.
- 4. Since there is no radiation in this process, it is a safest method among the other methods.

Limitations:

- 1. It is difficult to find the defects of the specimen which has complex shapes.
- 2. Trained, motivated technicians alone can perform this testing

8. With theory, explain the method of determination of velocity of ultrasonic waves in liquid using acoustic grating. (Jan 15, May 16)

PRINCIPLE:

Consider a container consisting of a liquid. An ultrasonic transducer is placed in the lower end of the container. If ultrasonic waves are passed through the liquid, alternate compression and rarefactions are produced. The compressed

portion acts as an opaque medium for the ordinary light and the rarefied portion acts as a transparent medium for the ordinary light. If the container is illuminated by a source of light, it undergoes diffraction and the diffraction pattern is obtained on a screen. So, the container acts as a grating. Therefore, it is said to be an acoustic grating (Fig.1.6)

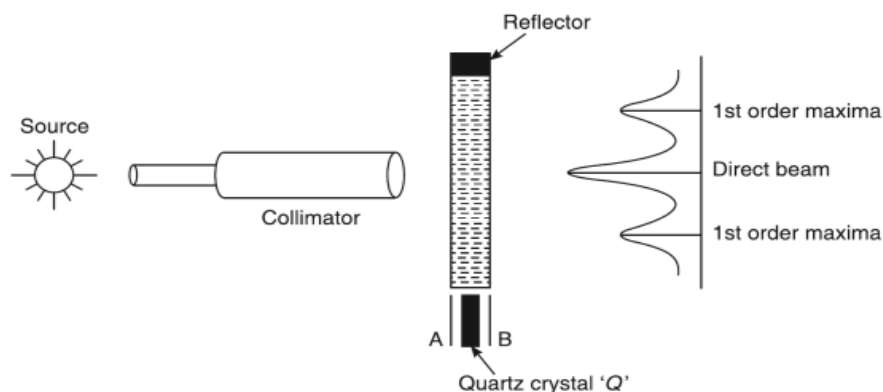


Fig.1.6. Acoustic grating

WORKING

(i) When the piezoelectric crystal is at rest:

When the quartz crystal is at rest, the light from the source is focused onto the glass tank filled with the liquid, a single image or vertical peak is observed on the screen. This shows that there is no diffraction of light.

(ii). When the piezoelectric crystal is set into vibration:

When ultrasonic waves are produced in the liquid by excitation of crystal, a number of

diffracted images appear on either side of the centre. The angular separation θ between the direct image of the slit and the diffracted image of any order, say $n = 1$, is measured.

Calculation of ultrasonic velocity

The velocity of ultrasonic waves can be determined using the condition.

$$2d \sin \theta = n\lambda \quad \dots(1)$$

$$\lambda = \frac{2d \sin \theta}{n} \dots \dots \dots (2)$$

$$\text{w.k.t } v = f \times \lambda$$

$$\lambda = \frac{v}{f} \quad \dots \dots \dots (3)$$

Substitute Eqn (2) in Eqn (1)

$$\frac{v}{f} = \frac{2d \sin \theta}{n}$$

$$v = 2d \sin \theta f \dots \dots \dots (4)$$

Where d is the distance between any two alternate compressions or rarefactions, θ is the angle of diffraction, n is the order to diffraction and λ is the wave length of the monochromatic source of light. By knowing the frequency f , d and the diffraction angle θ , the wavelength and the velocity of the ultrasonic wave is calculated from Eqn.(2 and 4).

9. What are the advantages and drawback of a Magnetostriction Oscillator? (Nov 16)

ADVANTAGES

- a) Oscillator construction is very simple and its production cost is low.
- b) At low frequencies larger power output is possible without the risk of damage to the oscillatory circuit

DISADVANTAGES

- c) It can produce frequencies up to 3 MHz only
- d) It is not possible to get a constant single frequency because it depends on the temperature and the degree of magnetization.
- e) Due to the presence of magnetic field, there will be loss of energy because of hysteresis and eddy current.

APPLICATION

Magnetostriction generator is used for high power applications such as drilling, grinding, welding etc.

10. Write a short note on NDT method? (May 14)

**Repeat above same answer (6&7) of i) PULSE ECHO METHODii)
LIQUID PENETRANT METHOD**

